

Smart parking sensors, technologies and applications for open parking lots: a review

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Abstract: Parking a vehicle in traffic dense environments often leads to excess time of driving in search for free space which leads to congestions and environmental pollution. Lack of guidance information to vacant parking spaces is one reason for inefficient parking behaviour. Smart parking sensors and technologies facilitate guidance of drivers to free parking spaces thereby improving parking efficiency. Currently, no such sensors or technologies is in use for open parking lot. This study reviews the literature on the usage of smart parking sensors, technologies, applications and evaluates their applicability to open parking lots. Magnetometers, ultrasonic sensors and machine vision were few of the widely used sensors and technologies on closed parking lots. However, this study suggests a combination of machine vision, convolutional neural network or multi-agent systems suitable for open parking lots due to less expenditure and resistance to varied environmental conditions. Few smart parking applications show drivers the location of common open parking lots. No application provided real-time parking occupancy information, which is a necessity to guide them along the shortest route to free space. To develop smart parking applications for open parking lots, further research is needed in the fields of deep learning and multi-agent systems.

1 Introduction

In densely populated areas as in cities, the availability of parking spaces is often less than the availability of vehicles which leads to a shortage of parking space [1]. Studies showed that in worldwide traffic dense environment, 30–50% of the drivers look for free parking space [2–5]. Based on previous studies, drivers use between 3.5 and 14 min to find a parking space [6, 7]. Examples of consequences are: the frustration of drivers, accidents, lost business opportunities, congestion and increased air pollution. Congestion can be observed when traffic density is increased, which is observed during peak traffic periods [8, 9]. A city with a population of ~50,000 inhabitants contains an average of 250 parking spaces in a parking lot which would generate an annual cost of ~216,000 US dollars [10].

In order to reduce parking related congestion and air pollution, public transportations such as bus, metro and so on can be utilised. However, private vehicles are still used for convenience. One way to address the carbon emission issue is to replace all fuel consumption vehicles with electric ones. Since, the average age of a car in Europe or the USA is ~11 years (USA) [11, 12], it can be expected to take quite a long time before electric vehicles replace all fuel consumption vehicles. In spite of using electric vehicles, cruising of a vehicle to find a vacant parking space would still be time consuming leading to congestion. A solution to that problem might be driverless cars. The driverless vehicle is one of the upcoming technologies currently being developed. Few companies such as Google have already developed prototype vehicles which support driverless mode in specific conditions [13]. With the current state of automation and proven technologies, new generation vehicles are providing park assist feature which helps in parking a car automatically with the use of sensors and guidance systems. Vehicle-to-vehicle (V2V) technology could be implemented on all vehicles in order to improve the safety of all driverless vehicles [14, 15]. An automated driverless vehicle would depend on a number of technologies such as global positioning system (GPS), sensors, V2V and so on and it would take few decades to replace existing vehicles. Yet another, not that far-fetched, solution might be a decision support system that uses smart parking sensor and technologies to detect parking occupancy information which facilitates the drivers to make informed decision

of where to go and park their car [16]. Parking occupancy information can be gathered using sensors and technologies such as ultrasonic sensors, magnetometers and multi-agent systems. Smart parking applications are available online which facilitate in improving parking efficiency by providing navigational directions to a reserved empty parking space [17].

There has already been a considerable amount of research conducted to improve parking efficiency at closed parking lots which are paid parking lots and support reservation of parking spaces. There are online applications which provide smart parking services in closed parking lots. However, literature is very scarce with regard to another and very common type of parking lot, the open parking lots which do not support reservation, freely available for a limited amount of time and are commonly placed outdoors occupying a large amount of space. Therefore, there still exists a research gap in order to improve parking efficiency at an open parking lot. It is assumed that the driver behaviour could be efficient if improved decision support to drivers is offered. The areal size of the often very large open parking lots could be reduced by improving parking efficiency along with reduced congestion and CO₂ emissions.

The aim of this paper is to review smart parking sensors, technologies and applications suitability at open parking lots. Challenges in an open parking lot are: freely available where the reservation is not possible and they are subjected to environmental conditions when placed outdoors. Therefore, suitability of sensors and technologies was made by compatibility and expenditure parameters.

The remaining part of this paper is organised in the following way. Section 2 describes the methods used to conduct the review of existing smart parking sensors, technologies and applications. Section 3 describes and analyses smart parking sensors, technologies and applications while Section 4 discusses about their suitability in open parking lots. Finally, the paper is ended with conclusion in Section 5.

2 Method

In this paper, literature search was carried out to identify articles related to smart parking sensors and technologies. Smart or intelligent parking has been in focus since early 1980s and there

are several papers discussing the implementation of sensors or technologies for improving parking efficiency. This paper will not review about parking requirements and income generated due to parking. Similarly, frameworks or architecture of smart parking systems are also not reviewed in this paper. This paper focuses on sensors, technologies and interfaces which are used to collect and display real-time parking occupancy information. The suitability of sensors and technologies is made in terms of compatibility and expenditure. Since, the majority of open parking lots are placed outdoors, compatibility is measured in terms of varying environment and lighting conditions while expenditure is measured in terms of purchase cost, installation and maintenance activities. Parking meters which are used for payments are placed either at a parking lot or beside parking spaces are not referred in this paper. From the literature search, literature reviews on sensors and technologies are presented in Table 1. However, it should be mentioned that these reviews do not emphasise on parking issues related to open parking lots.

Existing smart parking applications were searched using mobile application stores and web search engines. There are hundreds of parking applications available in the mobile application store. However, in this paper parking applications which provide real-time parking occupancy information and navigational directions to a reserved parking space are selected. All the referred applications support reservation which is possible only in closed parking lots. Smart parking applications also provide navigational directions to the reserved parking space in a closed parking lot.

3 Smart parking tools

Smart parking tools consist of sensors, technologies and applications which are used to identify parking occupancy information and facilitate to improve parking efficiency. Found literature reviews on smart parking technologies and sensors are shown in Table 1. In this context, literatures [18–20] are reviews about various types of smart parking sensors, technologies along with their uses while [21] identifies research gap in designing smart parking system for stakeholders. Furthermore, [22] reviews about intelligent parking technologies and their economic analysis and [23, 24] reviews the advantages and drawbacks of sensors and technologies. As shown in Table 1, machine vision using visual camera is referred by all the review articles. Sensors like ultrasonic and magnetometers are widely reviewed and tested which are used in various smart parking applications. Vehicular *ad hoc* networks (VANET) and fuzzy logic were not reviewed prior to 2013. A detailed description of each sensor and technology can be found in this section along with an emphasis on expenditure.

3.1 Smart parking sensors

There are various sensors which facilitate in detecting parking occupancy information and these are mentioned in the following sections. Sensors are one of the common tools which were widely

tested in several previous literatures. Descriptions of these sensors are mentioned in the following sections.

3.1.1 Passive infrared sensor: These sensors detect changes in energy and when a vehicle occupies a parking space, these sensors identify the change in energy and detect occupancy [25, 26]. The sensor observes a change in energy when a vehicle is placed or a person standing above the sensor. Based on the amount of energy change, it can be used to isolate outliers. However, these sensors are sensitive to environment and they would not be accurate when there is snow or rain. Passive infrared sensors should be placed under the ground or on the ceiling. Therefore, they require high investment for procurement and maintenance of these sensors. These sensors would be suitable for closed parking lots which are inside buildings and are not suitable for outdoor open parking lots.

3.1.2 Active infrared sensor: These sensors would emit infrared energy and detect any object or vehicle by the amount of energy reflected [25, 26]. They are also sensitive to environmental changes such as rain or snow. Therefore, they should be placed in all the parking spaces and require high investment and maintenance. Deploying sensors in all the parking spaces would help to attain parking occupancy status. These sensors are usually placed overhead and are suitable for indoor closed parking lots. As these sensors are sensitive to environmental changes, it is not suitable to open parking lots.

3.1.3 Ultrasonic sensor: These sensors would emit sound waves between 25 and 50 kHz and detect objects based on reflected energy. They are usually mounted on the ceiling and are sensitive to environmental changes such as rain and snow [27]. Therefore, they are suitable for indoor parking lots rather than open parking lots. Based on the distance at which waves are reflected it can distinguish between a vehicle and a person. In order to get parking occupancy status, these sensors should be placed on top of every parking space. These sensors would be available for low cost but installation and maintenance of multiple sensors and connecting them to a grid would be expensive in the long run. Wireless ultrasonic sensors are also used to gather parking occupancy information. They are connected using wireless sensor networks such as ZigBee protocol or other similar networks [28]. However, wireless sensors involve periodic maintenance costs. In another study, ultrasonic sensors are used on a drive-by vehicle and parking occupancy information is collected at regular periods [29]. Real-time parking occupancy information cannot be attained using the drive-by vehicle.

3.1.4 Inductive loop detectors: These detectors are installed using underground wiring system and they use principles of electromagnetism to detect the presence of a vehicle [25]. They are commonly used at the entrance and exit to know the count of the vehicles which can be used to know the availability of parking spaces. These detectors are expensive to install and maintain [26]

Table 1 Literature reviews on smart parking sensors and technologies

Smart parking tools	[19]	[24]	[18]	[22]	[21]	[20]	[23]
infrared sensors	•	•	•			•	•
ultrasonic sensors	•	•	•		•	•	•
inductive loop detectors	•	•	•				
parking guidance systems		•	•			•	
radio frequency tags		•	•		•	•	
magnetometer	•	•	•			•	•
microwave radar	•	•					
GPS		•	•	•	•		
machine vision	•	•	•	•	•	•	•
vehicular <i>ad hoc</i> networks				•	•		
multi-agent systems		•	•	•		•	
neural network					•		
fuzzy logic				•	•		

and they are commonly used in indoor parking lots to get the count of available parking spaces. An accurate count of vehicles would be provided using these detectors in a closed parking lot and these are in use at multiple commercial parking lots. However, individual parking occupancy status cannot be attained using inductive loop detectors.

3.1.5 Parking guidance systems: Parking guidance systems is another smart parking system which provides information about number of parking spaces available on display screens and these are usually placed near the parking lots as the driver can see and decide the parking space to occupy [28, 30]. Inductive loop detectors or visual camera can be used at the entrance and exit of a parking lot to know the count of the vehicles in a parking lot which would be displayed on the screens. However, they do not guide the driver to a particular parking space which is found empty. Therefore, there is every possibility that the driver would cruise for several minutes before finding an empty space to occupy. The driver can make a decision about the parking lot only after viewing the display screens [27]. Since sensors or visual cameras would be deployed only to get the count of vehicles, the expenditure for installation and maintenance would be minimal making them suitable for open parking lots.

3.1.6 Radio-frequency identification (RFID): Radio frequency tags are used to identify vehicle. Each vehicle will be given a radio frequency tag for identification. A transceiver and antenna would be installed at the entrance of a parking lot to identify the tag and allow the vehicle to occupy a parking lot [31]. These are suited for closed and indoor parking lots which are controlled. It is not suitable for an open parking lot as they are freely available. RFID is used to authorise movement of vehicles at a parking lot. However, it neither provides individual parking occupancy status nor facilitates the driver in finding a vacant parking space.

3.1.7 Magnetometer: These sensors detect the presence of vehicle by detecting the change in electromagnetic field. They need to be in close proximity to the vehicle, therefore, they are placed beneath the surface. They are not sensitive to the environment [25]. These are suitable for both open and closed parking lots. There are wireless sensors with a battery life time of few years which can be used to detect real-time parking occupancy information. The sensors should be placed under every parking space to know the occupancy of parking spaces. However, it is expensive to install and maintain these sensors on a large scale.

3.1.8 Microwave radar: A microwave radar transmits microwave beam and based on the reflected signal it estimates the velocity of the moving target. However, it does not detect stationary objects. In order to eliminate this restriction, dual microwave Doppler radar can be used to detect both moving and stationary vehicles [32]. These can be mounted or placed beneath the surface for vehicle detection. These radars are not sensitive to environment and can be used in open and closed parking lots. They should be placed in every parking space to detect parking occupancy status making them expensive to install and maintain these microwave radars on a large scale.

3.2 Smart parking technologies

Sensor technologies are tools which facilitate the driver in occupying a vacant parking space and descriptions of these technologies can be found in the following sections below.

3.2.1 Global positioning system: GPS-based navigational directions are provided to the driver for occupying a vacant parking space. GPS will facilitate in finding the shortest/optimal route from the current location. However, GPS alone cannot gather occupancy information of parking spaces. In one study occupancy of parking spaces is estimated using historical occupancy information and navigational directions are provided using GPS to the estimated parking space [33]. The accuracy of the GPS with a single-

frequency receiver is ≤ 7.8 m. If a dual-frequency receiver is used, the accuracy is < 0.71 m. A normal parking space would be between 2.3 and 2.7 m and most of the smartphones are provided with single-frequency receiver which has higher error compared to dual-frequency receivers. Dual-frequency receivers are usually used for military products for greater accuracy. A GPS is also perceptible to errors when the signal is blocked due to tall towers, walls within a building or under the ground. Therefore, navigational directions using GPS will be prone to errors in a closed indoor parking lot. Usage of GPS is suited for outdoor open parking lots where there is less chance for signal blocking. Accuracy of the GPS signal is also dependent on availability of satellite.

3.2.2 Machine vision: A visual camera can be used for license plate recognition (LPR) or identifying parking lot occupancy using machine vision. The camera should be placed near the entrance of a closed parking lot for LPR. Based on the number of vehicles entered and exited it can help to get the count of vacant parking spaces. However, occupancy status of parking spaces cannot be attained using this system. Video processing of parking lot using a camera is not ideal as it requires continuous transfer of large bandwidths. Therefore, a video should be broken into images at regular intervals and frame rates to facilitate continuous monitoring of the parking lot [23]. For parking spaces occupancy detection, a camera can be installed overhead to a parking lot and relevant image detection algorithms can be used to segment vehicles and detect occupancy of parking space. There are publicly available datasets such as PKLot [34] which consists of large number of outdoor parking lot images. In one of the studies, support vector machine classifier produced 89% accuracy on their test dataset. Similar image classification algorithms such as logistic regression, Viola and Jones [35] also produced higher accuracy results [36]. A camera is suited for open parking lots as it can cover large number of parking spaces [37]. However, it is susceptible to limitations such as occlusion and shadow effects, distortion and lighting change. These limitations can be removed with the use of 3D scene information [38]. Since limited number of cameras can cover large number of parking spaces, the expenditure is considered minimal.

3.2.3 Vehicular ad hoc networks: This system uses wireless communication devices to provide services such as smart parking and anti-theft. Road side units (RSUs) would be widely placed across parking lots and vehicles should be installed with on-board units (OBU). A trusted authority will be responsible for registrations of OBU and RSUs [39]. Therefore, once a vehicle approaches the parking lot installed with RSUs, navigational information to the vacant parking space will be provided to OBU. The OBU's can also be used by the vehicles to avoid collision while driving. Such features are also incorporated in self-driving vehicles. These devices are not sensitive to environment and are suitable for closed and open parking lots. However, installation and maintenance of RSUs in the parking lot would be expensive. In order to achieve accurate parking occupancy data and navigational information, all the vehicles must install OBU. Parking occupancy data is prone to errors if there are vehicles without OBU are parked.

3.2.4 Multi-agent systems: These kinds of systems make use of multiple mediums such as sensors, mobile, algorithms, visual camera and so on. These systems are also capable of incorporating aspects such as user preference, importance and so on, in finding a vacant parking space for the driver. Multi-agent systems are considered as foundation for automation of smart parking systems. A user can select a parking space using a mobile or web application and based on the user importance and preference, a parking space will be selected. The user will also receive navigational information to reach the parking space. Java tools such as JaCaMo and environment such as CARtAgo can be used in the architecture [40]. Machine vision systems or VANETs can be used instead of using sensors. Usage of multiple systems is supported in this architecture. These systems are suitable for both open and closed parking lots. The expenditure would be dependent

on the usage of technology to identify occupancy status of parking spaces.

3.2.5 Neural networks: Neural network is a data processing system which is inspired by brain nervous system. Neural networks have evolved over the years and various types of neural networks were developed such as fuzzy, neural network, fluid neural network, feed forward and convolutional neural network. Neural networks can be combined with machine vision to achieve automation. Neural networks were used in efficient recognition of license plates in real-time videos [41, 42]. In one study, images from morning and night were taken separately to train the neural network and a two layered feed-forward network with hidden sigmoid is used to produce accurate results in detection of available parking spaces [43]. Deep learning is a branch of machine learning which uses neural networks in object detection and classification. There is another evolving technology such as convolutional neural networks which would take images as input and is more efficient in analysing images. In a recent study, convolutional neural networks were used along with machine vision to capture parking occupancy information efficiently [44]. Convolutional neural network was also used to facilitate a self-driving vehicle. The network was trained using a set of connected cameras which observe the road while driving. Using limited training, the vehicle with convolutional neural network was able to operate on parking lots and roadways without lane markings [45]. This technology would function as an efficient tool in data processing while it is not involved in real-time data capturing. Therefore, it is suitable for open and closed parking lots with minimal expenditure.

3.2.6 Fuzzy logic: Fuzzy logic is an approach which incorporates multivalued logic in evaluating. Fuzzy logic can be used to develop forecasting models based on sample data. Similar to neural networks, fuzzy logic can also be used in multi-agent systems. According to a study, a sample of parking spaces availability information for 5 days using machine vision was taken to predict the availability of parking spaces in future dates using fuzzy logic [46]. Fuzzy logic supports autonomy in providing information on the availability of parking spaces. The accuracy of forecast models would not be high without validating with real-time data. Therefore, combination of fuzzy logic models with machine vision

or sensor technologies would increase the accuracy of the overall system. These systems are suitable for both open and closed parking lots. The expenditure would be minimal if image processing is used along with fuzzy logic to estimate available parking spaces for the future as well as provide real-time availability of parking spaces. Since this technology is not involved in the real-time data capturing process, it can be used for closed and open parking lots with minimal expenditure.

Not all of the sensors and technologies are suitable for gathering real-time occupancy information of open parking lots as mentioned in Table 2. Even though sensors are widely used to acquire parking occupancy information, they would be expensive to install and maintain on a large number of parking spaces. Few sensors such as infrared and ultrasonic are sensitive to environment and are not suitable for open parking lots. Technologies such as machine vision, multi-agent systems are suitable for open parking lots to acquire parking occupancy information and GPS can be used to provide navigational directions.

3.3 Smart parking applications

There are already many free smart parking applications available online in web and mobile stores of Android or iOS. Previously, reservation of parking space was done by calling to the service provider and now with the current usage of internet and smartphones, these services are provided online using mobile and web applications. These applications serve as decision support systems for the driver in occupying a vacant parking space. For instance, if the application shows a particular parking lot of choice to be full, the driver can search for nearby parking lots with available parking spaces or choose another destination. In this way, smart parking applications serve as decision support systems in occupying available parking spaces. Smart parking tools improve efficiency based on the following three categories:

- Guide the driver to parking lot using display boards.
- Reserve and authorise the driver to a parking lot.
- Reserve and guide the driver to a specific parking space using navigational information.

The three categories mentioned earlier are the ways in which efficiency of parking is improved using existing smart parking

Table 2 Strength and drawbacks of smart parking sensors and technologies in open parking lots

Smart parking technology	Suitability on open parking lots	Strength	Drawback
active/passive infrared sensor	X	provides parking occupancy status	not suitable for open parking lot due to varying environmental conditions
ultrasonic sensor	X	provides parking occupancy status	not suitable for open parking lot due to varying environmental conditions
inductive loop detectors	✓	facilitates in getting the count of vehicles in a parking lot	not suitable for individual parking occupancy status
parking guidance systems	✓	guide driver to a parking lot based on count of available parking spaces	do not solve congestion while cruising for parking space
radio frequency tags	X	facilitates in authorising vehicles in a parking lot	not suitable for open parking lots and cannot provide parking occupancy information
magnetometer	✓	provides parking occupancy status	expensive to install and maintain on large scale
microwave radar	✓	provides parking occupancy status	expensive to install and maintain on large scale
GPS	✓	provides navigational directions to the parking space	does not provide parking occupancy status and is subjective to errors
machine vision	✓	provides parking occupancy status or facilitates in authorising vehicles	challenges with low lighting conditions
VANET	✓	provides parking occupancy status	all vehicles should be part of the network without which will be prone to errors
multi-agent systems	✓	provides parking occupancy status	challenges will be based on the technology used
neural networks	✓	facilitates efficient parking occupancy detection	is used only as a supporting tool in parking occupancy detection
fuzzy logic	✓	provides parking occupancy status based on historical data	do not provide real-time parking occupancy

tools. The level of efficiency varies with each category. The first category would improve the parking efficiency, however cruising for the empty parking space would still be involved. The second category reserves a parking space for the driver, however, the driver would still search for the reserved parking space manually which also involves some amount of cruising while searching for the location of reserved parking space. The third category provides improved efficiency compared to the two previous categories as it facilitates in reserving and guiding the driver using the navigational information to the parking space.

The first category to guide the driver to a parking lot was provided with the use of parking guidance systems. This parking system facilitates the driver in taking a decision to park the vehicle. It is suitable to open parking lots as well. The second category would use RFID or LPR technologies. If a parking space is reserved by the driver, then RFID or LPR would authorise the vehicle to park. These are common methods for staff parking in an organisation and are not applicable to open parking lots. The third category would use web applications or smartphones to reserve the parking space and guide the driver using navigational directions through GPS. Table 3 shows smart parking applications available online which provide smart parking services as mentioned in the third category. All the applications are provided in limited cities of mentioned countries.

All the applications mentioned in Table 3 provide smart parking services in closed parking lots which support reservation and all of them are available in mobile application stores. However, Open Spot is the only application which is currently unavailable and discontinued by Google. Navigational directions to the free available parking space are provided which improve the efficiency of parking behaviour. A percentage or number of available parking spaces are shown for selected parking lots. None of the applications give the driver a choice in selecting a particular parking space. Most of the applications use underground wireless sensors such as magnetometers to get real-time parking occupancy information. Setup and maintenance cost of wireless sensors will be high and it is mandatory to replace all sensors again when the battery life is depleted. Since, a sensor needs to be placed in all the parking spaces, operational and maintenance costs of such sensors would be high. Crowdsourcing application is suited for open/closed parking lots and the expenditure is considered very minimal as no hardware installations or maintenance is required. However, the users need to update occupancy details of parking spaces every

time they park the vehicle and the accuracy of the information will be dependent on the user updating the occupancy details and is prone to human errors. Applications such as Parkopedia [47] and EasyPark Group [48] use historical data for predicting parking occupancy information. Algorithms like fuzzy logic, time series are used to predict parking occupancy information based on historical or sample data. These applications are operational in more number of cities and countries compared to other applications. All these applications provide an overview of parking occupancy which facilitates in decision making.

4 Discussion

The review of existing smart parking applications shows that most of these applications use sensors for parking occupancy detection in closed parking lots which would require a considerable amount of expenditure for installation and maintenance activities. AppyParking [49] application mentioned in Table 3 provides the location of open parking lots. However, parking occupancy information of open parking lot is not provided. Few applications such as EasyPark Group and Parkopedia use predictive analytics or crowdsourcing which can also be used on open parking lots. The applications which used predictive analytics were operational in more number of cities and countries than the applications which used sensors for parking occupancy detection. The difference might be due to less expenditure in using predictive analytics than deploying sensors in all parking spaces. However, real-time occupancy information of parking spaces cannot be acquired using predictive analytics and accuracy of parking occupancy information by crowdsourcing is not reliable. None of the applications provide real-time parking occupancy information of open parking lots. Lack of economic returns can be one of the reasons for not providing real-time occupancy information of open parking lots. Since there would not be any immediate economic gains with the use of smart parking tools at open parking lots, it would be well suited to use a sensor or technology with minimal expenditure to make it financially viable. A smart parking application would assist the driver as a decision support system and can also help in reducing congestion. However, using a smart parking application is optional and few drivers might not be convenient in using an application for parking occupancy. Complete benefits in terms of parking occupancy, land usage, congestion and pollution are only possible if every driver is using the application which is not practically possible.

According to previous literatures, several smart parking sensors and technologies were tested and reviewed. However, sensors or technologies were largely used for closed parking lots which support reservation. None of the technologies were used to improve the efficiency of parking in open parking lots. Ultrasonic and infrared sensors are sensitive to environmental conditions which might lead to inaccurate parking occupancy information. Sensors such as magnetometers and microwave radar are not sensitive to environmental conditions but are expensive to install and maintain at open parking lots. Even if the prices of sensors come down there would still be expenditure involved with installation and maintenance activities which would make it expensive to use sensors in open parking lots. Therefore, sensors are not ideal for open parking lots as expenditure should be minimal. Technologies such as VANET are also expensive and can be ruled out for open parking lots. However, VANET can be suitable in the future when all the vehicles are connected using on-board and road side units.

The parking occupancy detection using machine vision will vary between closed and open parking lots as there can be varying light conditions in an open outdoor parking lot. As mentioned earlier, challenges faced by low lighting and shadow conditions can be addressed. Therefore, machine vision using visual camera is one of the feasible smart parking technology for acquiring real-time parking occupancy information in open parking lots. Machine vision can be associated with convolutional neural networks and deep learning for efficient image classification. In the literature, neural networks are used mostly for LPR and there are limited studies which use convolutional neural networks for parking

Table 3 Smart parking applications and their use of technologies and sensors

Smart parking application	Country	Sensors/technology used
Park.ME	Austria, Germany	underground sensors
SmartParking	New Zealand	underground sensors, RFID
ParkMe	Japan, USA, UK, Germany, Brazil	underground sensors
ParkAssist	USA	M4 Smart sensors, LPR
SpotHero	USA	underground sensors
EasyPark	Canada	underground sensors
PaybyPhone	France	underground sensors
ParkMobile	USA	underground sensors
AppyParking	UK	magnetometer
EasyPark Group	Sweden, Denmark and so on	transactional data and crowdsourcing
Parker	USA	underground sensors, machine vision
Parkifi	USA	magnetometer
Best Parking	USA	underground sensors
Parkopedia	USA, Germany, Sweden and so on	predictive analytics, underground sensors
SFPark	USA	underground sensors
Open Spot	USA	crowdsourcing

occupancy detection. However, image classification accuracy is high while using convolutional neural networks.

In deep learning, large datasets such as Alexnet [50] are already available and additional data would facilitate in further improving image classification. Different technologies or image classification algorithms can be tested using PKLot which is publicly available outdoor parking lot images dataset [34, 36]. However, this dataset is not suitable if an infrared or thermal camera is used for data collection. Convolutional neural networks and deep learning are contemporary technologies which can be used in parking occupancy detection of open parking lots. As expenditure plays an important role in the choice of technology, it is found that machine vision, multi-agent systems, along with convolutional neural network are suitable to provide parking occupancy information. Data collected using the visual camera would be processed using image classification algorithms or convolutional neural network to generate parking occupancy information. This information can be displayed using smart parking applications along with navigational directions. Smart parking applications serve as decision support systems for the user as it helps the user to find a suitable parking space which is true in a closed parking lot. However, in an open parking lot, it might not be the case as a parking space selected by the user can be occupied by another vehicle. Therefore, in an open parking lot, a parking space should be updated in real time based on the availability and navigational directions should be provided to the updated parking space.

Along with real-time parking occupancy information, a short-term prediction of occupancy can be acquired using algorithms like fuzzy logic and time series. Since, reservation is not possible in an open parking lot, a short-term prediction would facilitate the driver in the decision-making process of choosing a parking lot. The prediction of parking space availability can also be made based on the location of the driver and its distance to the desired parking lot or destination. An optimal route can be calculated from the location to destination based on the distance of the route using which estimated arrival time can be acquired. Therefore, parking space occupancy prediction based on estimated arrival time can be shown to the driver. In this way, smart parking applications can serve as efficient decision support systems. Due to various types of parking lots, such as outdoor, multi-storey, basement or indoor, a combination of smart parking technologies and sensors can be used to improve the efficiency of parking behaviour with minimal expenditure.

Algorithms such as logistic regression, radial basis function, support vector machines, decision tree, random forest, Viola and Jones object detectors and so on can be used for pattern recognition and image classification. Support vector machines can be used to enhance feature extraction which would improve image classification. In methods such as convolutional neural networks, sample test and train images should be collected to train the network. The quality of classification would mostly depend on the trained sample images. The sample images consist of different vehicles and parking spaces at different environmental and light conditions. These images can be used to train convolutional neural network for efficient image classification. Usage of the visual camera for gathering parking occupancy information would also depend on a country's privacy protection laws. It can be used in public places with less restrictions in countries such as the UK and the USA. However, in countries such as Sweden, usage of surveillance camera is restricted unless it is used for preventing or investigating crimes [51]. A surveillance camera can be used without any restrictions if it is used in a private place without the involvement of public. In Sweden, the surveillance camera is restricted to use in an open parking based on privacy protection laws. Therefore, usage of surveillance camera's or similar technology might be subjected to individual countries regulations. In order to comply with the regulations, it is suggested to use a thermal camera as public cannot be recognised using it. Thermal cameras are expensive compared to a colour cameras. Since, either a colour- or thermal-based cameras can be used to cover a large number of parking spaces, the expenditure involved in using a thermal camera might not be high compared to the usage of

underground sensors. Future research can be made in using a thermal camera for parking occupancy detection.

Vehicle system technologies are moving towards automation and there are proven technologies such as park assist that are already available with new generation vehicles. Park assist feature will only help to park the vehicle without human intervention but it does not eradicate congestion. Such features can be improved if the vehicle is provided with real-time parking occupancy information. Therefore, parking occupancy information of an open parking lot can be gathered using a financially viable method which can be connected to automated driving vehicles for efficient parking. The usage of smart parking technology or sensor for parking occupancy information can be changed based on the kind of parking lot. If a parking lot is indoor and has less number of parking spaces, sensors such as magnetometers can also be used to collect parking occupancy information. If a parking lot is outdoor, the visual camera can be used at an elevated position to cover the entire parking lot. When a parking space is updated the navigational directions to enter the parking lot might also be updated and this might be a challenge to address when the user is already following navigational directions to a selected parking space. Future research can also be made to address this challenge.

5 Conclusion

This paper identifies a research gap in utilising smart parking sensors, technologies and applications for open parking lots. All the existing smart parking technologies and applications are not suitable for open parking lots due to varying environmental conditions and high expenditure. As there are no immediate economic gains from providing smart parking services in an open parking lot, expenditure plays an important role in the choice of smart parking technologies. Parking guidance system which is one of the existing smart parking technology can be used to get the count of available parking spaces in open parking lots. Machine vision is another technology which uses the visual camera to acquire real-time parking occupancy information on open parking lots due to its minimal expenditure. The usage of the visual camera is dependent on regulations supported by the country which needs to be considered prior. However, there is no single ideal technology suitable for parking occupancy detection. Based on the type of parking lot and size, a different combination of smart parking technologies and sensors can be used for efficient and financially viable parking occupancy detection. In order to further improve parking efficiency, navigational directions should be provided to a vacant parking space. Therefore, in order to address this challenge further research in the use of deep learning and multi-agent systems would help to provide real-time parking occupancy information along with navigational directions to available parking space in an open parking lot.

6 References

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